Engineering Physics (2025)
Course code 25PY101
Module 2 Unit 1: Quantum theories of solids

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Quantum Free Electron Theory

Fermi-Dirac distribution

Electronic specific heat of solids

Density of states (qualitative)

Success and Failures of quantum free electron theory of solids

E-k diagram

Classification of materials based on bands in solids

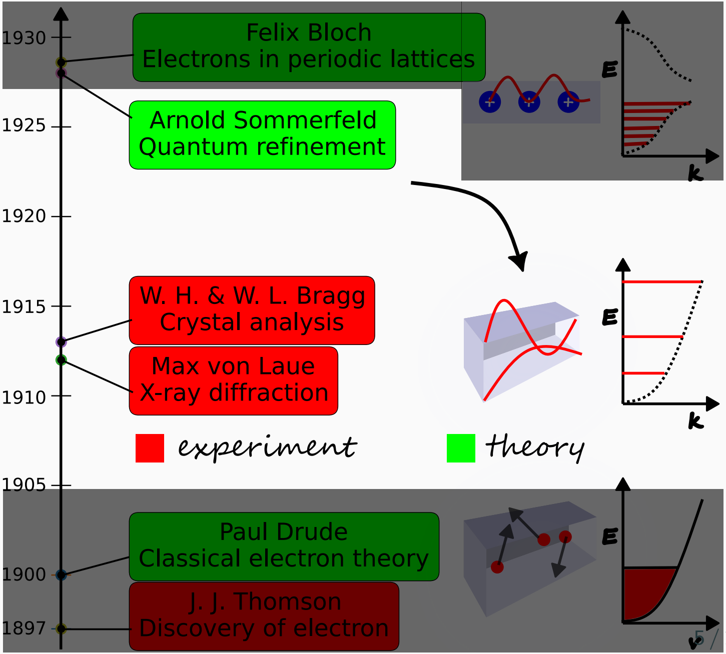
Fermi level in semiconductors- intrinsic and extrinsic

Summary of Classical free electron theory (CFET)

- Derived $\sigma = \dfrac{ne^2\tau}{m}$.
- Connected microscopic electron properties to macroscopic Ohm's law.
- Classical free electron theory has qualitative and quantitative agreement with experiment values of conductivity.
- However, the theory has drawbacks. Some of them are
    - Incorrect prediction for conductivity vs valency.
    - Cannot explain anomalous sign of Hall coefficient in some metals.
    - Underestimation of mean free path.
    - Cannot explain classification of materials into conductors. semi-conductors and insulators.
    - Wrong prediction of conductivity vs temperature.
    - Overestimation of heat capacity.

Electron theories of metals

1. Classical free electron theory
2. Quantum free electron theory
3. Quantum band theory



W. Pauli

Postulate

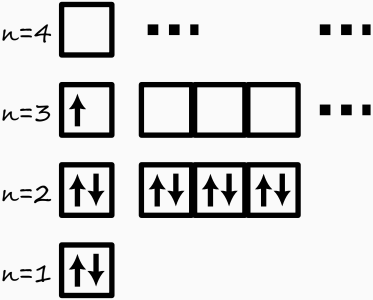
- No two electrons can occupy the same quantum state.
- No two electrons can share the same set of quantum numbers.
- Electron obeys Pauli's exclusion principle since it has spin $\frac{1}{2}$.

Definition

A quantum state is defined by a set of quantum numbers.

Problem

What quantum state is occupied by valence electron in Na?



Postulates

1. **Waves:** Electrons are quantum waves with wavevector k, angular frequency ω.
2. **Fermion:** Electron is a spin $\frac{1}{2}$ particle and obeys Pauli's exclusion principle.
3. **Independent electron approximation:** Electrons are independent and mutual repulsion between them is ignored.
4. **Free electron approximation:** Electrons are free and move in an infinite potential well.
5. **Quantum Thermodynamics:** The thermalization is governed by Fermi-Dirac statistics.

Density of states $Z(E) = \frac{d(\frac{N}{V})}{dE}$

Definition

- Density of states $Z(E)$ [1] is defined as the rate of change of the number of states per unit volume upto energy E with respect to energy E.
- Therefore, the the number of states per unit volume from energy E to $E + \mathrm{d}E$ is given by

$$Z(E)\,\mathrm{d}E.$$

Number of states per unit volume is

$$\rho_N(E) = \frac{N}{V} = \frac{\pi}{3}\left(\frac{\sqrt{8m}}{h}\right)^3 E^{3/2}$$

[1] In Neamen, density of states is denoted by $g(E)$.

DOS in textbook form

Differentiate $\rho_N(E)$ with respect to E to get the density of states function:

$$Z(E) = \frac{\mathrm{d}\rho_N(E)}{\mathrm{d}E} = \frac{\pi}{2} \cdot \left(\frac{\sqrt{8m}}{h}\right)^3 E^{1/2}$$

$$Z(E) = \frac{\pi}{2} \cdot 8 \cdot \left(\frac{2m}{h^2}\right)^{3/2} E^{1/2}$$

$$\boxed{\therefore Z(E) = 4\pi \cdot \left(\tfrac{2m}{h^2}\right)^{3/2} E^{1/2}.}$$

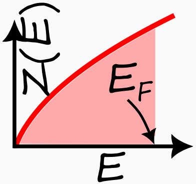
Key Insight
For 3D infinite potential, $Z(E) \propto \sqrt{E}$.

Problem
Find the form of $Z(E)$ for 2D infinite potential.

Fermi level

Definition
Fermi level of a metal is the maximum energy that an electron can have at $T = 0\,\mathrm{K}$.

At absolute zero all states are filled up to the Fermi energy E_F. The total electron density n_c (electrons per unit volume) is



$$
\begin{aligned}
n_c = \frac{N}{V} &= \int_0^{E_F} Z(E)\,\mathrm{d}E = 4\pi \cdot \left(\frac{2m}{h^2}\right)^{3/2} \int_0^{E_F} E^{1/2}\,\mathrm{d}E \\
&= 4\pi \cdot \left(\frac{2m}{h^2}\right)^{3/2} \cdot \frac{2}{3} E_F^{3/2} \\
&= \frac{8\pi}{3} \left(\frac{2m}{h^2}\right)^{3/2} E_F^{3/2}.
\end{aligned}
$$

Relating Fermi energy to electron density

Invert the relation to get the Fermi energy as a function of density:

$$E_F^{3/2} = \left(\frac{h^2}{2m}\right)^{3/2} \cdot \frac{3n_c}{8\pi}.$$

$$\boxed{\therefore E_F = \frac{h^2}{2m} \left(\frac{3n_c}{8\pi}\right)^{2/3}.}$$

This is the Fermi energy of a free electron gas in three dimensions at $T = 0$.

Problem
Calculate E_F for metal with electron density $n_c = 5.8 \times 10^{28}\,\mathrm{m}^{-3}$.

Test Your Understanding

1. What are the postulates of QFET.
2. Define Pauli's exclusion principle.
3. Define Fermi energy level.
4. Define density of states function.

Technical terms

- Pauli's exclusion principle
- Fermi level
- Density of states
- Fermi velocity
- Quantum state
- energy level
- degeneracy
- multiplicity

M2U1 Plan

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- Describes occupancy of energy states by electrons that obey **Pauli exclusion principle**.
- The probability of occupation of an energy level E at temperature T and Fermi energy level E_F is given by

$$\boxed{P_{FD}(E) = \frac{1}{\exp\left(\frac{E-E_F}{k_B T}\right)+1}.}$$

  Here k_B is the Boltzmann constant equal to $1.38 \times 10^{-23}\,\mathrm{J\,K^{-1}}$.
- To understand the FD statistics, let us analyze the probability function in two cases –
  1. $T \to 0$ limit
  2. $T \neq 0$

Key Insight
The Fermi-Dirac function is a result of Pauli exclusion principle.

Fermi–Dirac Statistics at the $T \to 0$ limit

- Take limit $T \to 0$ in definition:

$$P_{FD}(E) = \lim_{T\to 0} \frac{1}{\exp\left(\frac{E-E_F}{k_B T}\right)+1} = \frac{1}{\exp\left(\frac{E-E_F}{0}\right)+1}.$$

- This limit needs to analyzed for three cases –
  1. $E < E_F$
  2. $E > E_F$
  3. $E = E_F$

1. $E < E_F$

$$\begin{aligned} P_{FD}(E) &= \frac{1}{\exp\left(\frac{\text{-ve}}{0}\right)+1} \\ &= \frac{1}{\exp(-\infty)+1} \\ &= \frac{1}{0+1} \\ &= \frac{1}{1} \\ &= 1 \end{aligned}$$

2. $E > E_F$

$$\begin{aligned} P_{FD}(E) &= \frac{1}{\exp\left(\frac{+\text{ve}}{0}\right)+1} \\ &= \frac{1}{\exp(\infty)+1} \\ &= \frac{1}{\infty+1} \\ &= \frac{1}{\infty} \\ &= 0 \end{aligned}$$

3. $E = E_F$

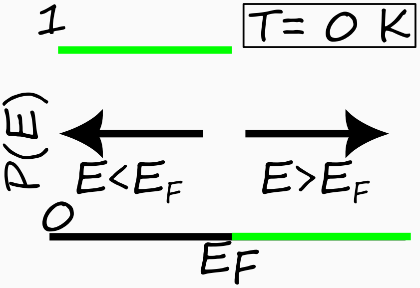
$$P_{FD}(E) = \frac{1}{\exp\left(\frac{0}{0}\right)+1}$$

- Since $\frac{0}{0}$ is indeterminate form, $P_{FD}(E)$ is not defined at $T = 0\,\mathrm{K}, E = E_F$.

- Hence, at $T = 0\,\mathrm{K}$ the distribution becomes a step function

$$P_{FD}(E) = \begin{cases} 1, & E < E_F \\ 0, & E > E_F \end{cases}$$

where E_F is the Fermi energy.



Key Insight
At $T = 0\,\mathrm{K}$, the Fermi-Dirac distribution is a step function.

Fermi–Dirac Statistics at $T \neq 0$

- At $T \neq 0$, consider $k_B T$ is a small positive quantity $+$ve.

$$P_{FD}(E) = \frac{1}{\exp\left(\frac{E-E_F}{+\text{ve}}\right) + 1}.$$

- This limit needs to analyzed for three cases –[2]
  1. $E \ll E_F$ so that $E - E_F$ is a big negative number i.e. --ve
  2. $E \gg E_F$ so that $E - E_F$ is a big positive number i.e. $++$ve
  3. $E = E_F$

[2] $\gg$ is symbol for very greater than and $\ll$ is symbol for very lesser than

Fermi–Dirac Statistics at $T \neq 0$

1. $E \ll E_F$ [a]

$$
\begin{aligned}
P_{FD}(E) &= \frac{1}{\exp\left(\frac{\text{- -ve}}{+ve}\right)+1} \\
&\simeq \frac{1}{\exp(-\infty)+1} \\
&= \frac{1}{0+1} \\
&= \frac{1}{1} \\
&= 1
\end{aligned}
$$

$\therefore P_{FD}(E) \simeq 1$

2. $E \gg E_F$

$$
\begin{aligned}
P_{FD}(E) &= \frac{1}{\exp\left(\frac{\text{++ve}}{+ve}\right)+1} \\
&\simeq \frac{1}{\exp(\infty)+1} \\
&= \frac{1}{\infty+1} \\
&= \frac{1}{\infty} \\
&= 0
\end{aligned}
$$

$\therefore P_{FD}(E) \simeq 0$

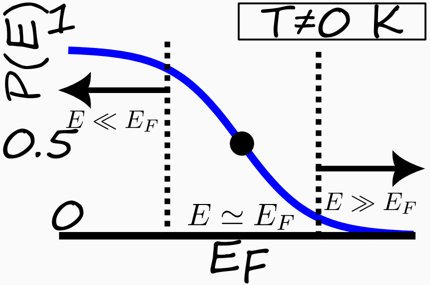
3. $E = E_F$

$$
\begin{aligned}
P_{FD}(E) &= \frac{1}{\exp\left(\frac{0}{+ve}\right)+1} \\
&= \frac{1}{1+1} \\
&= \frac{1}{2}
\end{aligned}
$$

[a] $\simeq$ is symbol for nearly equal.

- Hence, at $T \neq 0\,\mathrm{K}$ the distribution becomes a "smeared" [3] step function

$$P_{FD}(E) = \begin{cases} \simeq 1 & E \ll E_F \\ \simeq 0 & E \gg E_F \end{cases}$$



Key Insight
At $T = 0\,\mathrm{K}$, the Fermi-Dirac distribution is a "smeared" step function.

[3] smear means smoothened.

Problem
Calculate the probability that an energy level $3k_B T$ above the Fermi energy is occupied by an electron

Problem
Calculate the probability that an energy level $3k_B T$ below the Fermi level is empty

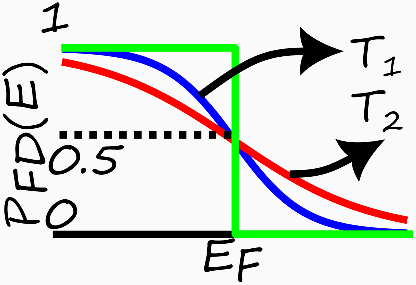
Problem
Fermi energy of a metal is $6.25\,\mathrm{eV}$. Calculate the temperature at which there is a $1\,\%$ probability that a state $0.30\,\mathrm{eV}$ below the Fermi energy level will not contain an electron.

Problem
Calculate the temperature at which the probability that an energy level $0.3\,\mathrm{eV}$ below the Fermi level is occupied is 0.7.

Problem

Calculate the temperature at which the probability that an energy level $3k_BT$ above the Fermi level is occupied is 0.7. *[Hint: This is a wrong question]*

Problem



Statement: $T_1 > T_2$.

Is the statement true or false?

Test Your Understanding

1. Explain the variation of Fermi-Dirac distribution function at $T = 0\,\mathrm{K}$ and $T \neq 0\,\mathrm{K}$.
2. Show that the probability of finding an electron of energy ΔE above the Fermi level is same as the probability of not finding an electron at energy ΔE below the Fermi level.
3. Show that the occupancy probabilities of two states whose energies are equally spaced above and below the Fermi energy add up to one.

Technical terms

- occupied level
- unoccupied level
- probability of occupancy
- probability of vacancy

M2U1 Plan

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Specific heat capacity

Definition

- Specific heat capacity [4] C is defined as the rate of change of energy E with temperature T per unit mole of material.

$$C = \frac{\mathrm{d}E}{\mathrm{d}T} \qquad \text{SI unit} \quad [C] = \left[\frac{E}{T} \cdot \frac{1}{\text{mole}}\right] = \mathrm{J\,K^{-1}\,mol^{-1}}.$$

- Metals heat "quickly". When we heat a metal, the heat energy is transferred to electrons. The quickness is measured in terms of heat capacity.

Problem
Specific heat of mercury is $0.14\,\mathrm{J\,g^{-1}\,K^{-1}}$, *water is* $1\,\mathrm{cal\,g^{-1}\,{}^\circ C^{-1}}$, *ethanol is* $2.44\,\mathrm{J\,g^{-1}\,{}^\circ C^{-1}}$. *Which has more specific heat? Which better material for thermometer?* [${}^{200}_{80}Hg$, $1\,\mathrm{cal}=4.2\,\mathrm{J}$, *ethanol* $= CH_3CH_2OH$]

[4] It is also called simply as specific heat.

Specific heat capacity – CFET

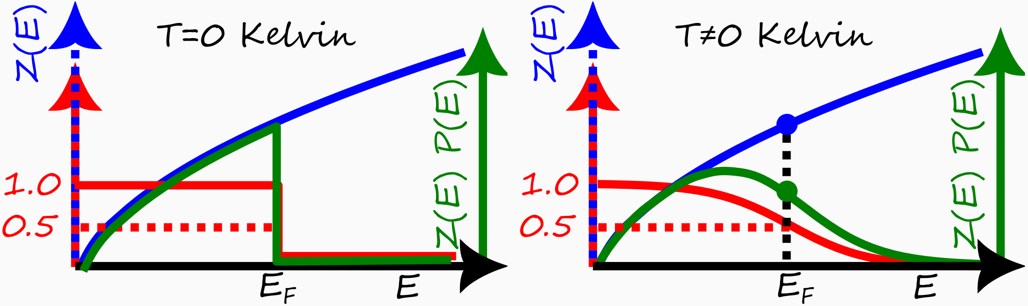
- Energy of electron at temperature T is $E_{\mathrm{el}} = \frac{3}{2} k_B T$.
- Energy of a mole of electrons is

$$E = E_{\mathrm{el}} N_A = \frac{3}{2} k_B N_A T = \frac{3}{2} RT. \qquad \left[R = 8.314\,\mathrm{J\,mol^{-1}\,K^{-1}}\right]$$

- Therefore, specific heat capacity is

$$C_{\mathrm{theory}}^{\text{classical}} = \frac{\mathrm{d}E}{\mathrm{d}T} = \frac{3}{2} R \sim 12\,\mathrm{J\,mol^{-1}\,K^{-1}}.$$

- But experimental values of specific heat capacity $C_{\text{experiment}}$ are **smaller by two orders** of magnitude.



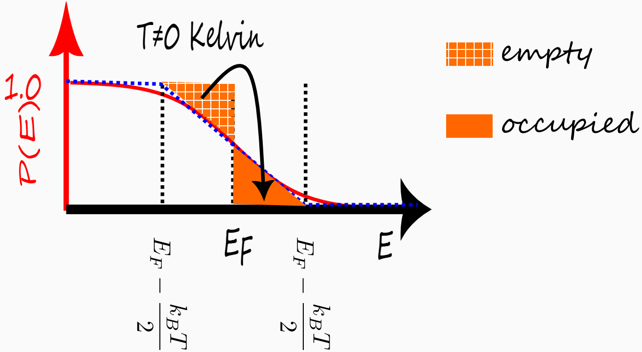
- For the derivation of specific heat capacity, only the **form** of density of states functions is necessary and not the exact expression. For 3D metal, $Z(E) \propto \sqrt{E}$ so that

$$Z(E) = \alpha\sqrt{E}, \qquad \text{where } \alpha \text{ is a constant}$$

- At $T = 0\,\mathrm{K}$, electron has energy less than or equal to Fermi energy level.
- The number of electrons per unit volume n_c at $T = 0\,\mathrm{K}$ is given by

$$
\begin{aligned}
n_c &= \int_0^{E_F} Z(E)P(E)\,\mathrm{d}E \\
&= \int_0^{E_F} Z(E)\,\mathrm{d}E \\
&= \int_0^{E_F} \alpha\sqrt{E}\,\mathrm{d}E \\
&= \frac{2}{3}\alpha E_F^{3/2}
\end{aligned}
$$

$$\boxed{\therefore\ n_c = \tfrac{2}{3}\alpha E_F^{3/2}.}$$



- The number of thermalized electrons is equal to area of the filled triangle multiplied by the density of states in the neighbourhood.
- Area of the filled triangle is

$$A = \frac{1}{2}\frac{k_B T}{2}\frac{1}{2} = \frac{1}{8}k_B T$$

$$\therefore\ n_{\text{thermal}} = \frac{1}{8}k_B T \cdot Z(E_F) = \frac{1}{8}\alpha k_B T\sqrt{E}$$

Specific heat capacity – Energy of thermalized electron

- The net increase in energy of thermalized electron is

$$\Delta E_{\text{thermalized}} = E_F + \frac{k_B T}{2} - \left(E_F - \frac{k_B T}{2} \right) = k_B T$$

- Since only a fraction of the electrons are thermalized, the net increase in energy of 1 electron is

$$\begin{aligned} \Delta E_{1\text{ el}} &= \Delta E_{\text{thermalized}} \cdot f_{\text{thermalized}} \\ &= \frac{3}{16} k_B T \cdot \frac{k_B T}{E_F} \end{aligned}$$

- Therefore, the net increase in energy of 1 mole of electrons is

$$\begin{aligned} \Delta E_{1\text{ mole}} &= \Delta E_{1\text{ el}} \cdot N_A \\ &= \frac{3}{16} (k_B N_A) T \cdot \frac{k_B T}{E_F} \\ &= \frac{3}{16} R \frac{k_B T^2}{E_F} \end{aligned}$$

- The specific heat capacity is given by

$$
\begin{aligned}
C_{\text{theory}}^{\text{quantum}} &= \frac{\mathrm{d}E_{1\ \text{el}}}{\mathrm{d}T} \\
&= \frac{3}{8} R \frac{k_B T}{E_F} \\
&= \frac{3}{2} R \cdot \left(\frac{1}{4} \frac{k_B T}{E_F} \right) \\
&= C_{\text{theory}}^{\text{classical}} \cdot \left(\frac{1}{4} \frac{k_B T}{E_F} \right)
\end{aligned}
$$

$$
\boxed{\therefore\ C_{\text{theory}}^{\text{quantum}} \simeq \left(\frac{k_B T}{E_F} \right) \cdot C_{\text{theory}}^{\text{classical}}.}
$$

- Therefore, quantum theory corrects the classical expression by the fraction $\frac{k_B T}{E_F}$. At room temperature, specific heat capacity is given by

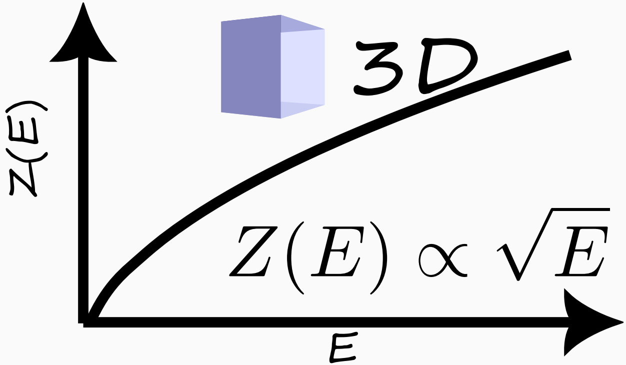
$$
C_{\text{theory}}^{\text{quantum}} = C_{\text{theory}}^{\text{classical}} \cdot \frac{k_B T_{300\,\text{K}}}{E_F} \simeq 0.01\, C_{\text{theory}}^{\text{classical}} \simeq C_{\text{experiment}}
$$

This validates Quantum Free Electron Theory.

M2U1 Plan

Density of states (3D)

- The form of the DOS in 3D is $Z(E) \propto \sqrt{E}$



Problem

If $Z_{3D}(E) = k\sqrt{E}$, where k is a constant, find the expression for conduction electron number density n_c in terms of Fermi level E_F at $T = 0\,\mathrm{K}$ in a 3D material.

M2U1 Plan

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Merits and demerits of QFET

Merits

- Electrical conductivity
- Thermal conductivity
- Heat capacity

Demerits

- Cannot explain classification of condensed matter into metals, semiconductors and insulators
- Occurence of positive Hall coefficient in some metals like Zn.

$E - k$ diagram – 1D case

- In a 1D metal, the electron is free as well as independent.
- For a free electron, the energy is related to the momentum p by

$$E = \frac{p^2}{2m}.$$

- However, since the electron behaves also as a wave, its wavelength λ is related to the momentum by the de Broglie relation

$$\lambda = \frac{h}{p} \quad \Rightarrow \quad p = \frac{h}{\lambda}.$$

- By definition, the wavevector k of the electron is inversely related to the wavelength as

$$k = \frac{2\pi}{\lambda}$$

- Therefore, the energy is related to the wavevector by

$$E = \frac{1}{2m}\left(\frac{h}{\lambda}\right)^2 = \frac{1}{2m}\left(\frac{hk}{2\pi}\right)^2 = \frac{\hbar^2 k^2}{2m}.$$

Summary of Quantum free electron theory (QFET)

- Derived density of states function $Z(E)$.
- Quantum free electron theory addressed the electronic contribution to specific heat of metals
- However, the theory has drawbacks. Some of them are
  - Cannot explain anomalous sign of Hall coefficient in some metals.
  - Cannot explain classification of materials into conductors. semi-conductors and insulators.

Reason for drawbacks

- Invalidity of free electron approximation.
- Drawbacks addressed by Bloch's quantum band theory.

Test Your Understanding

1. If the Fermi level of metal is E_f, what is the fraction of thermalized electrons at temperature T.
2. A thermalized electron according to CFET increases the energy of metal by $\mathcal{O}(k_B T)$. However, according to QFET, the thermalized electron increases the energy of metal by $\mathcal{O}(fk_B T)$, where f is the fraction of thermalized electrons. Explain.
3. Prove that the dispersion relation $E - k$ in metals is given by

$$E(k) = \frac{\hbar^2 k^2}{2m}.$$

4. List the merits and demerits of QFET.

Technical terms

- specific heat capacity C
- ideal gas constant R
- thermalized electron
- $E - k$ dispersion relation

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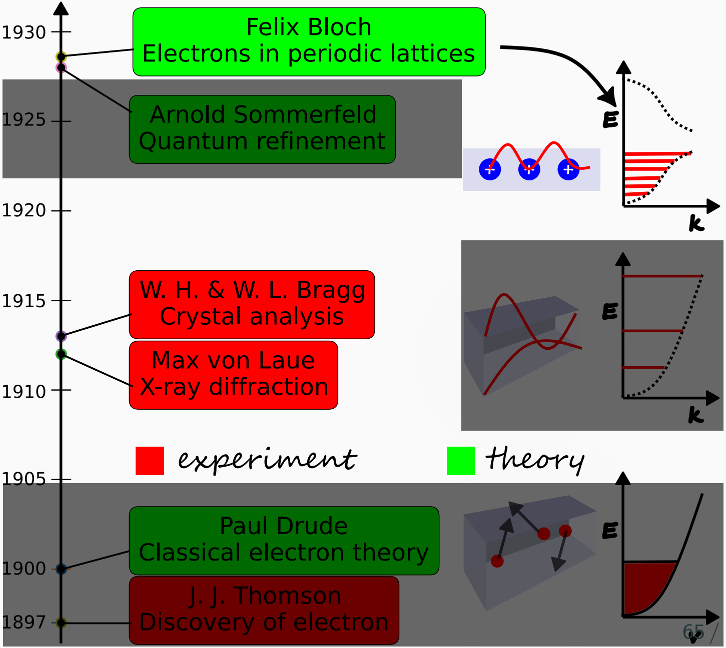
Electron theories of metals

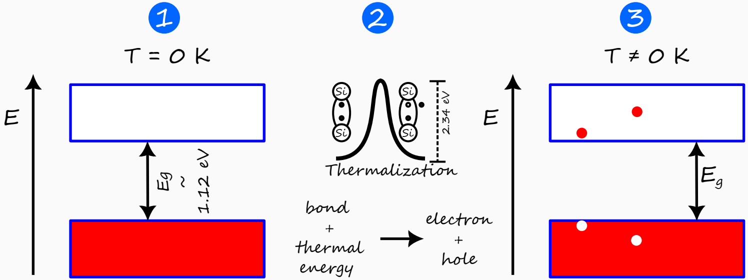
1. Classical free electron theory
2. Quantum free electron theory
3. Quantum band theory (QBT)



Postulates

1. **Waves:** Electrons are quantum waves with wavevector k, angular frequency ω.
2. **Fermion:** Electron is a spin $\frac{1}{2}$ particle and obeys Pauli's exclusion principle.
3. **Independent electron approximation:** Electrons are independent and mutual repulsion between them is ignored.
4. **Periodic potential approximation:** Electrons move in a periodic potential.
5. **Quantum Thermodynamics:** The thermalization is governed by Fermi-Dirac statistics.

Energy band diagram



1. At $T = 0\,\mathrm{K}$,
   - The topmost occupied band is called **valence band**.
   - The lower most unoccupied is called **conduction band**.
   - These two bands are separated by a forbidden gap called the **energy band gap**.
2. When temperature is increased, the covalent bond is broken due to thermalization. This results in the formation of **electron-hole pair** [5].
3. At $T \neq 0\,\mathrm{K}$, the electron occupies energy level in the conduction band whereas the hole occupies energy level in the valence band.

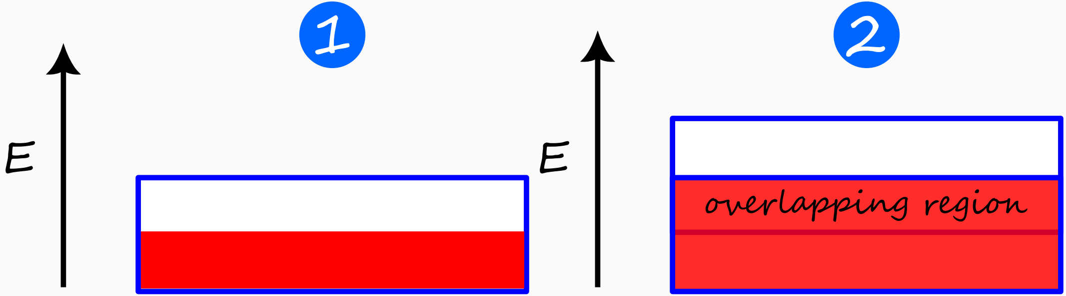
[5] Hole is the absence of electron in the valence band.

Classification of solids

- According to band theory, conductivity of solid is determined by the band gap separating the conduction band and valence band.
- Based on the band gap, the solids are classified into
    1. metals or conductors
    2. semiconductors
    3. insulators

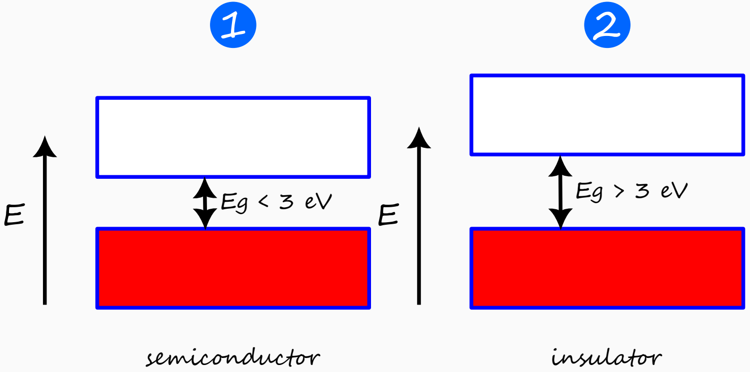
Key Insight
The classification of solids is based on energy band gap.

Energy band diagram – Conductor



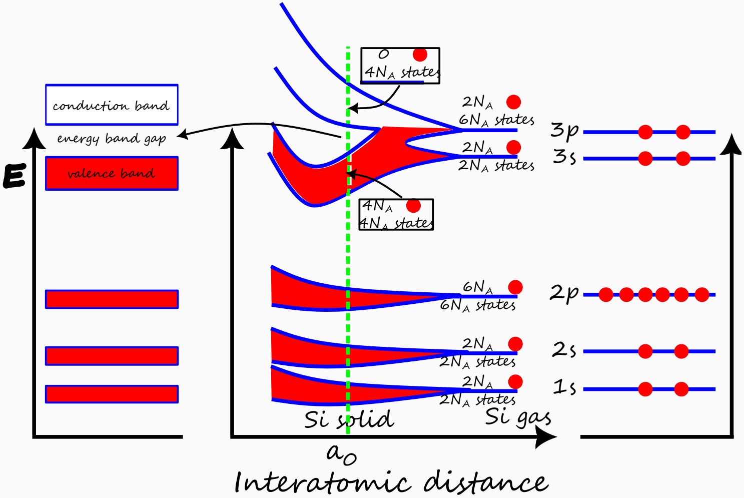
- In conductors, there are two cases
  1. the conduction band is half filled.
  2. the conduction band overlaps with empty upper band.
- In both cases, there is no band gap so that empty states are available for conduction.
- Since there is no band gap, there is no absence of electron at $T \neq 0$. Therefore, there are no holes in conductors.

Energy band diagram – Semiconductor



- In either semiconductor or insulator, there is a band gap so that empty states are not available for conduction.
  1. The energy band gap is less than 3 eV for semiconductor.
  2. The energy band gap is greater than 3 eV for insulator

Energy band diagram – Si



Key Insight

The valence band is completely filled and the conduction band is completely empty at $T = 0\,\mathrm{K}$.

Test Your Understanding

1. What are the postulates of QBT?
2. Define the periodic potential approximation.
3. In the parabolic approximation, since the conduction band is approximated by an upward parabola, the conduction band charge carrier is electron-like. Explain.
4. Hole has a negative effective mass. Explain.

Technical terms

- periodic potential approximation
- forbidden energy level
- band structure
- conduction band
- valence band
- energy band gap
- energy band diagram
- conduction band minimum E_c
- valence band maximum E_v
- parabolic approximation
- effective mass of electron and hole

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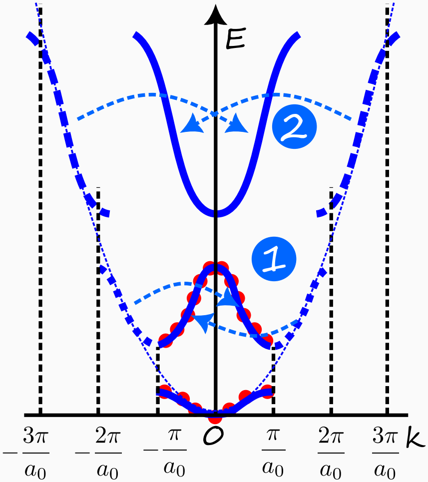
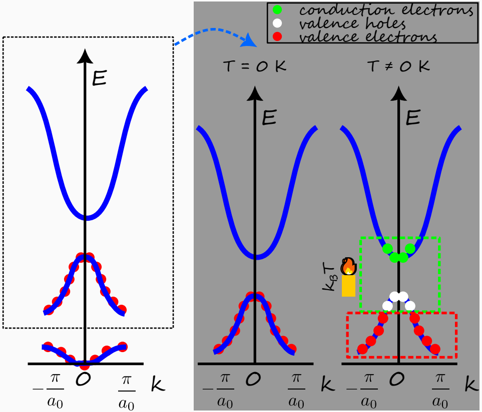
Reduced band dispersion – Band structure



- Due to the periodicity of lattice, the band dispersion can be analyzed within the **zone** from $k = -\frac{\pi}{a_0}$ to $k = +\frac{\pi}{a_0}$.
- The original band dispersion can be reduced to the zone by folding the "arms" of the parabola resulting in the reduced band dispersion. This process is called **zone folding**.
- The reduced band dispersion is called the **band structure** of the material and it is the signature of the material.

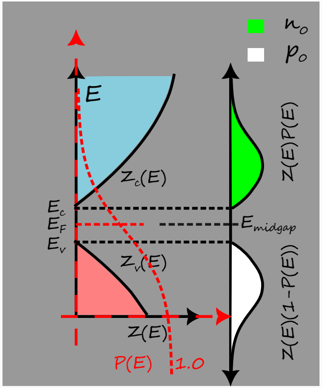
Key Insight
Bandstructure is the fingerprint or "DNA" of a crystalline solid.



- For a semiconductor, due to the presence of energy band gap, there are no free electrons at $T = 0\,\mathrm{K}$.
- At $T \neq 0\,\mathrm{K}$, the electrons near the valence band maximum thermalize and jump to the levels near the conduction band minimum. This results in quantum vacancy in the valence band. This is called **hole**.
- Conduction electron and valence hole contribute to conduction.

Key Insight

- Hole is the absence of electron in the valence band.
- Hole is not a physical reality.



- The conduction electron density n_0 is given by

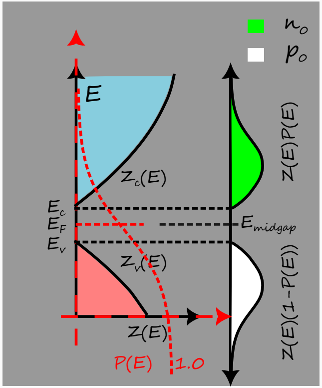
$$n_0 = \int_{E_c}^{\infty} Z_c(E) P(E)\, \mathrm{d}E$$
$$= \int_{E_c}^{\infty} \frac{\alpha^* \sqrt{E - E_c}}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)} \mathrm{d}E$$

- By applying Boltzmann approximation and the exact expression for α^*, it can be shown that

$$\boxed{n_0 = N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right)}$$

where N_c is the **effective density of states** in the conduction band.

Hole carrier concentration



- Similarly the valence hole density p_0 is given by

$$p_0 = \int_{E_c}^{\infty} Z_c(E)\,(1 - P(E))\,\mathrm{d}E$$

$$= \int_{0}^{E_v} \beta^* \sqrt{E_v - E}\left(1 - \frac{1}{1 + \exp\left(\frac{E - E_F}{k_B T}\right)}\right)\mathrm{d}E$$

- By applying **Boltzmann approximation** and the exact expression for β^*, it can be shown that

$$\boxed{p_0 = N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right)}$$

where N_v is the **effective density of states** in the valence band.

Intrinsic carrier concentration – $n_0 p_0$ product

- For an intrinsic semiconductor, electron-hole pairs are generated at $T \neq 0\,\mathrm{K}$. So the conduction electron density is equal to valence hole density

$$n_0 = p_0 = n_i$$

where n_i is defined as the intrinsic carrier concentration.

- The intrinsic carrier concentration is given by the product of electron and hole concentrations

$$\begin{aligned}
n_0 p_0 = n_i^2 &= N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) \cdot N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right) \\
&= N_c N_v \exp\left(-\frac{E_c - E_F + E_F - E_v}{k_B T}\right) \\
n_i^2 &= N_c N_v \exp\left(-\frac{E_c - E_v}{k_B T}\right)
\end{aligned}$$

Intrinsic carrier concentration

- Therefore, the $n_0 p_0$ is independent of Fermi energy level and is given in terms of the material properties and temperature.

$$\boxed{n_0 p_0 = n_i^2}$$

where n_i is given by

$$\begin{aligned} n_i &= \sqrt{N_c N_v \exp\left(-\frac{E_c - E_v}{k_B T}\right)} \\ &= \sqrt{N_c N_v} \exp\left(-\frac{E_c - E_v}{2k_B T}\right) \\ &= \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right) \end{aligned}$$

where E_g is the energy band gap and is given by

$$E_g = E_c - E_v$$

$$\boxed{\therefore n_i = \sqrt{N_c N_v} \exp\left(-\frac{E_g}{2k_B T}\right)}$$

Fermi level of intrinsic semiconductor

- Since the electron and hole exist in pairs in an intrinsic semiconductor

$$\begin{aligned} n_0 &= p_0 \\ \Rightarrow N_c \exp\left(-\frac{E_c - E_F}{k_B T}\right) &= N_v \exp\left(-\frac{E_F - E_v}{k_B T}\right) \end{aligned}$$

- Taking logarithm on both sides of the equation, we have

$$\begin{aligned} \ln N_c + \ln\left[\exp\left(-\frac{E_c - E_F}{k_B T}\right)\right] &= \ln N_v \ln\left[\exp\left(-\frac{E_F - E_v}{k_B T}\right)\right] \\ \Rightarrow \ln N_c - \frac{E_c - E_F}{k_B T} &= \ln N_v - \frac{E_F - E_v}{k_B T} \\ \Rightarrow \frac{2E_F}{k_B T} &= \frac{E_c + E_v}{k_B T} + \ln N_v - \ln N_c \\ \Rightarrow E_F &= \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln \frac{N_v}{N_c} \end{aligned}$$

- If we substitute the expressions for N_c and N_v, in the ratio of N_c over N_v all the terms cancel out except the effective masses of electron and hole.

Fermi level of intrinsic semiconductor

- The expressions for effective density of states are

$$N_c = 2\left(\frac{2\pi m_n^* k_B T}{h^2}\right)^{3/2}, \qquad N_v = 2\left(\frac{2\pi m_p^* k_B T}{h^2}\right)^{3/2}$$

- Therefore, the Fermi level in terms of effective masses of charge carriers is given by

$$\begin{aligned} E_F &= \frac{E_c + E_v}{2} + \frac{k_B T}{2} \ln\left(\frac{m_p^*}{m_n^*}\right)^{3/2} \\ &= E_{\text{midgap}} + \frac{3k_B T}{4} \ln\frac{m_p^*}{m_n^*} \end{aligned}$$

where E_{midgap} is defined as the midgap energy level that is in the midpoint of conduction band minimum and valence band maximum and is given by

$$E_{\text{midgap}} := \frac{E_c + E_v}{2}$$

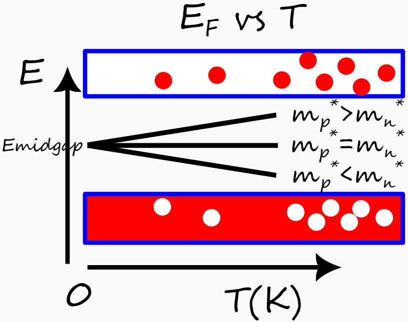
$$\boxed{\therefore E_F = E_{\text{midgap}} + \tfrac{3k_B T}{4} \ln \tfrac{m_p^*}{m_n^*}}$$

E_F vs T of intrinsic semiconductor

- Since the Fermi energy level varies with temperature, let us consider two cases – $T = 0\,\mathrm{K}$ and $T \neq 0\,\mathrm{K}$.
- Also, the Fermi energy level varies with effective masses of electron and hole. So let us consider three cases – $m_p = m_n$, $m_p > m_n$, and $m_p < m_n$.

	$T = 0\,\mathrm{K}$	$T \neq 0\,\mathrm{K}$
$m_p = m_n$	E_{midgap}	E_{midgap}
$m_p > m_n$	E_{midgap}	$E_{\text{midgap}} + \delta$
$m_p < m_n$	E_{midgap}	$E_{\text{midgap}} - \delta$

Table 1: Here δ is a positive quantity that linearly increases with temperature.



Key Insight
With temperature, E_F deviates from E_{midgap} for semiconductors with differing electron and hole effective masses.

Conductivity of intrinsic semiconductor

- Conduction in a semiconductor is through two channels – electron and hole. Therefore

$$\sigma_i = \sigma_e + \sigma_h$$

- For each channel, the conductivity is given by the product of charge carrier concentration and mobility. Therefore

$$\begin{aligned}\sigma_i &= n_0 e \mu_e + p_0 e \mu_h \\ &= n_i e \left(\mu_e + \mu_h\right)\end{aligned}$$

$$\boxed{\therefore \sigma_i = n_i e \left(\mu_e + \mu_h\right)}$$

- If we express the intrinsic carrier concentration in terms of material properties, then

$$\begin{aligned}\sigma_i &= \sqrt{N_c N_v} e \left(\mu_e + \mu_h\right) \exp\left(-\frac{E_g}{2k_B T}\right) \\ &= \sigma_0 \exp\left(-\frac{E_g}{2k_B T}\right)\end{aligned}$$

where σ_0 is a constant in terms of material properties.

Some useful information for calculations

	Si	GaAs	Ge
μ_n ($\mathrm{cm^2\,V^{-1}\,s^{-1}}$)	1350	8500	3900
μ_p ($\mathrm{cm^2\,V^{-1}\,s^{-1}}$)	480	400	1900

Table 2: Mobility of carriers in Si, GaAs, and Ge at 300 K. Taken from Section 5.1 of Neamen

	Si	GaAs	Ge
n_i ($\mathrm{cm^{-3}}$)	1.5×10^{10}	1.8×10^{6}	2.4×10^{13}

Table 3: Intrinsic carrier concentration in Si, GaAs, and Ge at 300 K.

	Si	GaAs	Ge
N_c (cm^{-3})	2.8×10^{19}	4.7×10^{17}	1.04×10^{19}
N_v (cm^{-3})	1.04×10^{19}	7.0×10^{18}	6.0×10^{18}
m_n^*/m_0	1.08	0.067	0.55
m_p^*/m_0	0.56	0.48	0.37

Table 4: Effective density of states and effective mass of charge carriers in Si, GaAs, and Ge . Taken from Section 4.1.3 of Neamen.

	Si	GaAs	Ge
E_g (eV)	1.12	1.42	0.66

Table 5: Energy band gap of Si, GaAs, and Ge at 300 K. Taken from Appendix. B of Neamen.

Problem
A quantum state in the conduction band is $k_B T$ above conduction band minimum E_c of an intrinsic semiconductor. The Fermi level is $0.25\,\mathrm{eV}$ below E_c. Find the probability that the quantum state is occupied at $300\,\mathrm{K}$.

Problem
For the same material in above problem, if the effective density of states in conduction band N_c is $2.8 \times 10^{19}\,\mathrm{cm}^{-3}$ at $300\,\mathrm{K}$, then find the electron concentration n_0.

Problem
Calculate the position of the intrinsic Fermi level with respect to the center of the bandgap in Si at $300\,\mathrm{K}$.

Problem
Find the resistivity of intrinsic Ge at $300\,\mathrm{K}$. *[Hint: Check useful info slide]*

Problem
The resistivity of intrinsic Si is $2.3 \times 10^3\,\Omega\,\mathrm{m}$ *at* $300\,\mathrm{K}$. *Calculate its resistivity at* $100\,^\circ\mathrm{C}$. *[Hint:* $E_g^{Si} = 1.12\,\mathrm{eV}$ *]*

Tough problems

Problem
Calculate the thermal equilibrium electron concentration in Si at $400\,\mathrm{K}$. N_c *at* $300\,\mathrm{K}$ *is* $2.8 \times 10^{19}\,\mathrm{cm}^{-3}$. *[Hint: Solve the below problem.]*

Problem
Calculate the ratio of conduction electrons at $400\,\mathrm{K}$ *to conduction electrons at* $300\,\mathrm{K}$.
[Hint: From the explicit expression for N_c, *observe the form of* N_c *vs* T. *]*

Test Your Understanding

1. Discuss the variation of Fermi level position with temperature.
2. Discuss the variation of conductivity with temperature.

Technical terms

- band structure
- zone folding
- hole
- conduction band effective density of states N_c
- conduction band effective density of states N_v
- intrinsic carrier concentration n_i
- midgap energy level E_{midgap}